Pulmonary Involvement

Most patients will present with symptoms such as **dyspnoea**, a **dry cough**, and **reduced exercise tolerance**. Chest X-ray can be useful but is a relatively insensitive method for the detection of interstitial lung disease (ILD) and is no longer recommended at the time of first diagnosis. **High-resolution computed tomography (HRCT)** of the chest has markedly higher diagnostic sensitivity and is the recommended imaging modality to assess the extent and distribution of ILD. The sensitivity of HRCT is superior compared to lung function testing (LFT).

LFT should include **spirometry**, **body plethysmography**, and measurement of the **diffusing capacity of the lung for carbon monoxide (DLCO)**, corrected for haemoglobin. These tests should be performed every 6 months, or more frequently if the patient shows a decline in forced vital capacity (FVC) and/or DLCO.

Pulmonary Hypertension

Pulmonary arterial hypertension (PAH) occurs in approximately 15% of patients with systemic sclerosis (SSc), particularly in those with long disease duration and **anticentromere antibodies**. PAH is associated with significant mortality and is among the leading causes of death in SSc.

All SSc patients should be evaluated for possible PAH in accordance with current guidelines and referred for specialist management when indicated. **Annual screening** is strongly recommended and should include assessment of symptoms—such as unexplained or progressive dyspnoea, syncope, and signs of right heart failure—as well as **echocardiography**. This approach is endorsed by major cardiologic and pulmonary societies (e.g., 2015 ESC Guidelines).

Recent pooled analyses of clinical trials and registries have shown substantially improved outcomes and survival in SSc-PAH due to advances in care and treatment. In addition to supporting the use of currently available therapies, evidence for the benefit of **combination therapy** offers a model for improving management across other organ systems affected by SSc.

Gastrointestinal Involvement

Gastrointestinal (GI) tract involvement is common in SSc, affecting up to **80%** of patients in the oesophagus and **40–70%** in the stomach, small intestine, and colon. In longstanding disease (i.e., >10 years), upper GI involvement is nearly universal.

Common manifestations include: - **Heartburn** and **oesophageal dysfunction** (due to impaired motility) - **Diarrhoea** secondary to bacterial overgrowth (from small intestinal dysmotility) - **Faecal incontinence** (from anorectal involvement)

Barrett's oesophagus may develop as a late complication of chronic gastro-oesophageal reflux and requires surveillance per established guidelines.

Rarely, **mucosal telangiectasias** may occur, which can be a source of occult gastrointestinal bleeding. The standard diagnostic procedure for evaluating these lesions is **endoscopy**.

Cardiac Involvement

Cardiac disease in SSc is influenced by the degree of **myocardial fibrosis**, as well as the added strain from concurrent **lung fibrosis** and **pulmonary vascular disease**. **Myocarditis** and **pericarditis** may occur in some patients and can complicate diagnosis.

Risk factors for cardiac involvement include: - **Diffuse cutaneous SSc** - Rapid disease progression - Signs of systemic inflammation (e.g., tendon friction rubs)

Patchy myocardial fibrosis leads to **diastolic dysfunction** and may reduce **left ventricular ejection fraction** over time.

Arrhythmias are common, and in patients with diffuse SSc, severe arrhythmias are a significant cause of mortality. Given the low sensitivity of standard electrocardiography, there should be a low threshold for performing **Holter monitoring** to detect arrhythmic events.

Renal Involvement

Scleroderma renal crisis (SRC) is a life-threatening complication characterized by **acute kidney injury**, **microangiopathic haemolytic anaemia (MAHA)**, and **accelerated phase hypertension**. It most commonly occurs in patients with **diffuse cutaneous SSc** within the first **4 years** of disease onset.

The presence of **anti-RNA polymerase III antibodies** is a major risk factor and is found in about one-third of SRC cases.

To enable early detection: - Blood pressure should be monitored **at least twice weekly**, preferably with home monitoring. - **Glucocorticoid use** should be minimized; doses exceeding **15 mg of prednisone equivalents per day** are associated with increased risk of renal crisis and should be avoided.